Set - A

CHAITANYA

(DEEMED TO BE UNIVERSITY)

II INTERNAL EXAMINATIONS–APRIL -2026

B.Tech(CSE) - II SEMESTER

NUMERICAL ABILITY

[Time: 1 ½ Hrs.] **[Max. Marks: 20]**

Section - A

Answer any six Questions. ***6 x 2 = 12***

1.	Answer the following Questions: a. How many odd days are there in 123 days? b. How many odd days are there in 366 days?
2.	If 1st January 2020 was a Wednesday, what will be the day of the week on 1st January 2021
3.	A small cafe recorded the number of orders over 3 days: ● Monday: 120 orders ● Tuesday: 150 orders ● Wednesday: 130 orders Out of total orders, 40% were online orders. What is the total number of online orders over the 3 days?
4.	In a queue of 50 people, A is 16th from the left and B is 19th from the right. How many people are there between A and B?
5.	Number of students enrolled in a course over years: ● 2018: 200 ● 2019: 240 ● 2020: 300 ● 2021: 360 What is the percentage increase in students from 2018 to 2020?
6.	Find the missing term in the series: 6, 18, 36, 60, ?, 126
7.	Ravi starts from his house and walks 10 m towards North. He then turns right and walks 5 m, and finally turns right again and walks 10 m. In which direction is Ravi now from his starting point?
8.	In a marathon event, participants finished the race in a sequence. Arjun finished 12th from the start, and there were 35 runners in total. What was Arjun's position from the end?
9.	If "STAR" is written as "RSZQ" in a certain code, then how will "MOON" be written in that code?
10.	Priya starts from her home and walks 12 m towards East. She then turns left and walks 5 m. After that, she turns left again and walks 12 m. How far is Priya from her starting point?

Section - B

Answer the following Questions. ***2 x 4 = 8***

11	Find the day of the week for 11th September 2002 using the odd days concept.
	OR
12	Starting from point P, Rohit walks 12 km towards the south. He then takes a right turn and walks 8 km, followed by another right turn where he walks 15 km. After that, he turns left and walks 6 km. (a) Calculate the total distance travelled by Rohit. (b) Find the shortest distance between point P and Rohit's final position, and also determine the direction of Rohit's final position with respect to point P.

13

Store	January Sales	February Sales
A	120	150
B	200	180
C	160	220
D	140	170
E	180	210

The table below shows the number of mobile phones sold by five stores (A, B, C, D, and E) in two different months.
Answer the following Questions:
(a) Which store recorded the highest total sales across both months?
(b) What is the difference in total sales between Store C and Store B?
(c) What is the average sales per store in February?
(d) What is the percentage increase in sales of Store A from January to February?

OR

14 Answer the following number series questions:
18, 41, 78, 129, ?, 273
(a) Identify the missing term.
(b) Find the next term after 273 using the same pattern.

Set - B

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Section - A

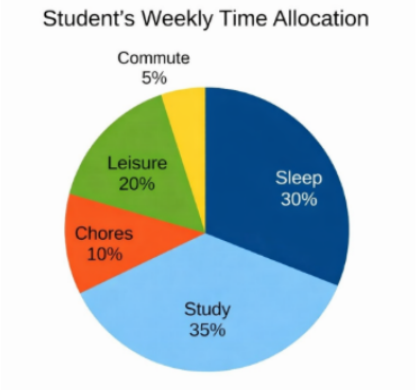
Answer any six Questions. ***6 x 2 = 12***

1.	Answer the following Questions: a. How many odd days are there in a non-leap year? b. How many odd days are there in a leap year?
2.	If today is Friday, what day will it be after 199 days?
3.	A fruit shop sold apples over 3 days: ● Monday: 30 apples ● Tuesday: 45 apples ● Wednesday: 25 apples What is the average number of apples sold per day?
4.	In a row of students, P is 14th from the left and Q is 11th from the right. If there are 25 students in total, what is the position of Q from the left?
5.	A mobile shop recorded its monthly sales of smartphones over the four months: ● January: 200 units ● February: 180 units ● March: 150 units ● April: 250 Units What is the percentage decrease in sales from January to March?
6.	Find the missing term in the series: ANB, CPD, ERF, ?
7.	Joseph starts from a point and walks 14 km towards the west. He then turns right and walks 9 km, turns right again and walks 14 km, and finally turns right and walks 5 km. How far is Joseph from his starting point, and in which direction is he located?
8.	Anitha is 8th from the top, and Deepak is 6th from the bottom. If 5 people are ranked between Anitha and Deepak, what is the total number of people in the group?
9.	In a certain code, ROMAN is written as "# \ @ % *" and SEA is written as "& ! %". How will "ROSE" be written in that code?
10.	A boy starts from home, walks 10 km south, turns right and walks 7 km, then turns right and walks 10 km, and finally turns left and walks 3 km. In which direction is he now from his home?

Section - B

Answer the following Questions. ***2 x 4 = 8***

11.	Find the day of the week for 18th September 2003 using the odd days concept.
	OR
12.	Ravi starts from his home at point P in the morning. He goes for a walk to explore the nearby area. He first walks 12 km towards the north. From there, he turns east and walks 5 km. Next, he moves 8 km towards the south, and finally walks 7 km towards the west. (a) What is the shortest distance between his final position and point P? (b) In which direction is he located from point P?

13.	 The pie chart shows how a student's weekly time is distributed among different activities. Note: Yellow Represents Commute - [5%] (a) Which activity consumes the highest percentage of the student's weekly time? (b) If there are 168 hours in a full week, how many hours does the student spend on leisure activities? (c) What is the ratio of time spent on study compared to time spent on chores? (d) If the student spends 56 hours per week on all activities, how many hours are spent on studying and sleeping together?
	OR
14.	In a class of 60 students, the number of girls is twice the number of boys. A boy named Arjun holds the 12th rank from the top. It is also known that 8 girls are ranked above him. (a) Determine the number of boys and girls in the class. (b) Find how many boys are ranked below Arjun. (c) What is Arjun's rank among the boys from the top?

Set - C

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NUMERICAL ABILITY

Time: 1 ½ Hrs.] [Max. Marks: 20

Section - A

Answer any six Questions. 6 x 2 = 12

1.	Answer the following Questions: a. Determine the number of odd days in 200 years. b. Determine the number of odd days in 300 years.
2.	Assuming that 1st April 2024 is a Monday, find all the dates in April 2024 that fall on a Thursday.
3.	A shop sold mobile phones over 4 months: • January: 120 phones (40% were Android phones) • February: 150 phones (50% were Android phones) • March: 180 phones (60% were Android phones) • April: 200 phones (55% were Android phones) How many Android phones were sold in total across all four months?
4.	In a class of 60 students, Ravi is ranked 15th from the top. Another student, Arjun, is ranked 42nd from the top. How many students are there between Ravi and Arjun in the rank list?
5.	A company's revenue across 4 years is as follows: • In Year 1, the revenue is ₹2,50,000 • In Year 2, the revenue is ₹3,00,000 • In Year 3, the revenue is ₹3,75,000 • In Year 4, the revenue is ₹4,50,000 What is the percentage increase in revenue from Year 1 to Year 4?
6.	Find the missing term in the series: 12, 15, 21, 30, ?, 57
7.	A person starts from point P and walks 10 km north. Then he turns 90° right and walks 6 km. After that, he turns 90° right again and walks 10 km. Finally, he turns 90° left and walks 4 km. What is the shortest distance between his final position and point P, and in which direction is he from point P?
8.	In a certain code, 2345 is written as MOPU and 1670 is written as RNST. How is 5071 written in that code?
9.	In a row of 80 students, Ravi is ranked 22nd from the left. Another student, Arjun, is ranked 15th from the right. How many students are there between Ravi and Arjun?
10.	Rahul starts from point A, walks 9 km north to B, then 12 km east to C, then 9 km south to D, and finally 12 km west to A. What total distance did Rahul cover?

Section - B

Answer the following Questions. 2 x 4 = 8

11.	If 1st January 2000 was a Saturday, Answer the following: (a) What day of the week was 1st January 2001? (b) How many total odd days are there from the year 2000 to 2003(Including 2000 & 2003). (c) What day of the week was 3rd January 2004?
	OR
12.	In a class of 90 students, Rahul is ranked 24th from the top. The number of students below Rahul is 3 times the number of students above him. Another student Arjun is ranked 18th from the bottom. (a) Find the total number of students in the class. (b) What is Rahul's rank from the bottom? (c) How many students are between Rahul and Arjun? (d) If 5 students above Arjun leave the class, what will be Arjun's new rank from the top?

13.

Brand	Q1	Q2	Q3	Q4
A	120	150	180	200
B	140	160	200	220
C	100	140	160	210
D	130	170	190	230
E	110	150	170	240

Table: Sales of Laptops by Five Brands (in units)

Answer the following Questions:

(a) Which brand shows the highest percentage growth in sales from Q1 to Q4?

(b) What is the percentage increase in total sales of all brands from Q1 to Q4?

OR

14. Two people, Ravi and Arjun, start from the same point.

- Ravi walks 12 km East and then 5 km North.
- Arjun walks 7 km West and then 5 km North.

(a) What is the distance between Ravi and Arjun?

(b) In which direction is Ravi with respect to Arjun?

(c) In which direction is Arjun located from the starting point?